

Assignment of Fundamentals of Financial Mathematics

Yichuan Lin

At 15:30 on December 09, 2024, the stock price for McDonald's Corporation (MCD), denoted as S_0 , is recorded at \$298.98 USD according to data from Yahoo Finance. The screenshots below illustrate the market prices for both call and put options set to mature on January 16, 2026, and the volatility we use.



301.29 +0.58 (+0.19%) **302.42** +1.71 (+0.57%)

At close: 9:30:51 AM EST Pre-Market: 9:29:55 AM EST

Dec 13, 2024 290.00 List All Options

Calls In The Money

Contract Name	Last Trade Date (EDT)	Expiry Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213C00290000	12/10/2024 7:59 PM	2024-12-13	12.51	0.00	0.00	0.00	0.00%	8	183	0.00%
MCD241220C00290000	12/10/2024 8:23 PM	2024-12-20	13.45	0.00	0.00	0.00	0.00%	18	1,066	0.00%
MCD241227C00290000	12/10/2024 8:45 PM	2024-12-27	13.15	0.00	0.00	0.00	0.00%	3	91	0.00%
MCD250103C00290000	12/10/2024 4:49 PM	2025-01-03	12.30	0.00	0.00	0.00	0.00%	1	66	0.00%
MCD250110C00290000	12/5/2024 2:51 PM	2025-01-10	11.10	0.00	0.00	0.00	0.00%	-	2	0.00%
MCD250117C00290000	12/10/2024 8:03 PM	2025-01-17	16.00	0.00	0.00	0.00	0.00%	87	1,944	0.00%
MCD250124C00290000	12/6/2024 5:49 PM	2025-01-24	15.37	0.00	0.00	0.00	0.00%	1	1	0.00%
MCD250222C00290000	12/10/2024 3:51 PM	2025-02-21	17.20	0.00	0.00	0.00	0.00%	1	175	0.00%
MCD250321C00290000	12/10/2024 8:47 PM	2025-03-21	20.72	0.00	0.00	0.00	0.00%	4	758	0.00%
MCD250620C00290000	12/10/2024 8:43 PM	2025-06-20	25.95	0.00	0.00	0.00	0.00%	8	316	0.00%
MCD250918C00290000	12/10/2024 7:45 PM	2025-09-18	30.50	0.00	0.00	0.00	0.00%	2	41	0.00%
MCD260116C00290000	12/10/2024 8:44 PM	2026-01-16	36.45	0.00	0.00	0.00	0.00%	2	879	0.00%
MCD261218C00290000	12/9/2024 3:08 PM	2026-12-18	44.87	0.00	0.00	0.00	0.00%	1	1,124	0.00%
MCD270115C00290000	12/5/2024 2:43 PM	2027-01-15	43.50	0.00	0.00	0.00	0.00%	1	37	0.00%

Puts In The Money

Contract Name	Last Trade Date (EDT)	Expiry Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213P00290000	12/10/2024 8:34 PM	2024-12-13	0.14	0.00	0.00	0.00	0.00%	202	789	12.50%
MCD241220P00290000	12/10/2024 8:33 PM	2024-12-20	0.50	0.00	0.00	0.00	0.00%	127	2,188	6.25%
MCD241227P00290000	12/10/2024 7:24 PM	2024-12-27	0.85	0.00	0.00	0.00	0.00%	4	178	3.13%
MCD250103P00290000	12/10/2024 8:49 PM	2025-01-03	1.32	0.00	0.00	0.00	0.00%	10	554	3.13%
MCD250110P00290000	12/10/2024 5:28 PM	2025-01-10	1.91	0.00	0.00	0.00	0.00%	11	37	3.13%
MCD250117P00290000	12/10/2024 8:53 PM	2025-01-17	2.35	0.00	0.00	0.00	0.00%	163	1,527	3.13%
MCD250221P00290000	12/10/2024 6:44 PM	2025-02-21	4.85	0.00	0.00	0.00	0.00%	74	345	1.56%
MCD250321P00290000	12/10/2024 8:00 PM	2025-03-21	6.15	0.00	0.00	0.00	0.00%	45	850	1.56%
MCD250620P00290000	12/9/2024 3:43 PM	2025-06-20	9.82	0.00	0.00	0.00	0.00%	1	236	1.56%
MCD250918P00290000	12/9/2024 3:27 PM	2025-09-18	13.20	0.00	0.00	0.00	0.00%	1	157	0.78%
MCD260116P00290000	12/6/2024 4:01 PM	2026-01-16	15.75	0.00	0.00	0.00	0.00%	5	261	0.78%
MCD261218P00290000	12/6/2024 3:04 PM	2026-12-18	22.60	0.00	0.00	0.00	0.00%	1	71	0.78%
MCD270115P00290000	11/27/2024 2:33 PM	2027-01-15	25.12	0.00	0.00	0.00	0.00%	1	46	0.78%

McDonald's Corporation (MCD) ★ Following ↕ Compare

301.76 -1.05 (-0.35%) **302.42** +1.71 (+0.57%)

At close: 9:30:12 AM EST Pre-Market: 9:29:55 AM EST

Dec 13, 2024 280.00 List All Options

Calls In The Money

Contract Name	Last Trade Date (EDT)	Expiry Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213C00290000	12/5/2024 6:31 PM	2024-12-13	16.00	0.00	0.00	0.00	0.00%	1	5	0.00%
MCD241220C00290000	12/9/2024 5:39 PM	2024-12-20	23.20	0.00	0.00	0.00	0.00%	2	136	0.00%
MCD241227C00290000	12/9/2024 5:54 PM	2024-12-27	20.00	0.00	0.00	0.00	0.00%	2	8	0.00%
MCD250103C00290000	12/6/2024 3:37 PM	2025-01-03	23.15	0.00	0.00	0.00	0.00%	1	88	0.00%
MCD250110C00290000	12/3/2024 8:43 PM	2025-01-10	17.20	0.00	0.00	0.00	0.00%	-	104	0.00%
MCD250117C00290000	12/10/2024 7:24 PM	2025-01-17	23.75	0.00	0.00	0.00	0.00%	12	1,861	0.00%
MCD250221C00290000	12/9/2024 5:06 PM	2025-02-21	27.00	0.00	0.00	0.00	0.00%	7	59	0.00%
MCD250321C00290000	12/10/2024 8:23 PM	2025-03-21	28.10	0.00	0.00	0.00	0.00%	18	819	0.00%
MCD250620C00290000	12/10/2024 8:10 PM	2025-06-20	33.23	0.00	0.00	0.00	0.00%	21	911	0.00%
MCD250918C00290000	12/10/2024 5:02 PM	2025-09-18	35.29	0.00	0.00	0.00	0.00%	2	177	0.00%
MCD260116C00290000	12/10/2024 6:02 PM	2026-01-16	38.50	0.00	0.00	0.00	0.00%	2	671	0.00%
MCD261218C00290000	11/23/2024 3:48 PM	2026-12-18	43.34	0.00	0.00	0.00	0.00%	1	5	0.00%
MCD270115C00290000	12/5/2024 6:02 PM	2027-01-15	50.00	0.00	0.00	0.00	0.00%	3	38	0.00%

Puts In The Money

Contract Name	Last Trade Date (EDT)	Expiry Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213P00290000	12/10/2024 8:29 PM	2024-12-13	0.04	0.00	0.00	0.00	0.00%	291	2,274	12.50%
MCD241220P00290000	12/10/2024 8:55 PM	2024-12-20	0.16	0.00	0.00	0.00	0.00%	39	1,933	12.50%
MCD241227P00290000	12/9/2024 8:13 PM	2024-12-27	0.32	0.00	0.00	0.00	0.00%	24	171	6.25%
MCD250103P00290000	12/10/2024 8:32 PM	2025-01-03	0.40	0.00	0.00	0.00	0.00%	8	39	6.25%
MCD250110P00290000	12/10/2024 4:22 PM	2025-01-10	0.89	0.00	0.00	0.00	0.00%	10	55	6.25%
MCD250117P00290000	12/10/2024 8:42 PM	2025-01-17	0.91	0.00	0.00	0.00	0.00%	241	10,885	6.25%
MCD250221P00290000	12/10/2024 4:42 PM	2025-02-21	1.83	0.00	0.00	0.00	0.00%	2	3	6.25%
MCD250321P00290000	12/10/2024 3:53 PM	2025-03-21	3.48	0.00	0.00	0.00	0.00%	7	171	3.13%
MCD250620P00290000	12/10/2024 8:32 PM	2025-06-20	3.73	0.00	0.00	0.00	0.00%	4	901	3.13%
MCD250918P00290000	12/10/2024 5:43 PM	2025-09-18	7.10	0.00	0.00	0.00	0.00%	9	1,078	1.56%
MCD260116P00290000	12/10/2024 2:34 PM	2026-01-16	10.40	0.00	0.00	0.00	0.00%	15	123	1.56%
MCD260116P00290000	12/10/2024 2:45 PM	2026-01-16	14.05	0.00	0.00	0.00	0.00%	8	161	1.56%
MCD261218P00290000	12/5/2024 5:20 PM	2026-12-18	20.80	0.00	0.00	0.00	0.00%	1	68	0.78%
MCD270115P00290000	12/6/2024 6:23 PM	2027-01-15	20.04	0.00	0.00	0.00	0.00%	1	22	0.78%

301.29 +0.58 (+0.19%) **302.42** +1.71 (+0.57%)

At close: 9:30:13 AM EST Pre-Market: 9:29:55 AM EST

Dec 13, 2024 295.00 List All Options

Calls In The Money

Contract Name	Last Trade Date (EDT)	Expiry Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213C00290000	12/10/2024 8:50 PM	2024-12-13	6.60	0.00	0.00	0.00	0.00%	47	311	0.00%
MCD241220C00290000	12/10/2024 8:54 PM	2024-12-20	7.68	0.00	0.00	0.00	0.00%	169	2,253	0.00%
MCD241227C00290000	12/10/2024 7:17 PM	2024-12-27	6.80	0.00	0.00	0.00	0.00%	47	351	0.00%
MCD250103C00290000	12/10/2024 8:48 PM	2025-01-03	9.42	0.00	0.00	0.00	0.00%	41	240	0.00%
MCD250110C00290000	12/10/2024 4:26 PM	2025-01-10	9.00	0.00	0.00	0.00	0.00%	2	25	0.00%
MCD250117C00290000	12/10/2024 8:55 PM	2025-01-17	11.30	0.00	0.00	0.00	0.00%	55	3,414	0.00%
MCD250221C00290000	12/10/2024 8:18 PM	2025-02-21	12.92	0.00	0.00	0.00	0.00%	25	27	0.00%
MCD250321C00290000	12/10/2024 8:27 PM	2025-03-21	16.65	0.00	0.00	0.00	0.00%	10	340	0.00%
MCD250620C00290000	12/10/2024 8:43 PM	2025-06-20	17.30	0.00	0.00	0.00	0.00%	43	490	0.00%
MCD250918C00290000	12/10/2024 7:31 PM	2025-09-18	22.60	0.00	0.00	0.00	0.00%	8	525	0.00%
MCD260116P00290000	12/5/2024 8:56 PM	2026-01-16	25.55	0.00	0.00	0.00	0.00%	2	148	0.00%
MCD261218C00290000	12/10/2024 8:53 PM	2026-12-18	31.38	0.00	0.00	0.00	0.00%	7	193	0.00%
MCD270115P00290000	12/9/2024 4:00 PM	2026-12-18	43.00	0.00	0.00	0.00	0.00%	1	31	0.00%

Puts In The Money

Contract Name	Last Trade Date (EDT)	Expiry Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213P00290000	12/10/2024 8:58 PM	2024-12-13	0.47	0.00	0.00	0.00	0.00%	247	308	6.25%
MCD241220P00290000	12/10/2024 8:38 PM	2024-12-20	1.15	0.00	0.00	0.00	0.00%	99	1,294	3.13%
MCD241227P00290000	12/10/2024 8:22 PM	2024-12-27	1.48	0.00	0.00	0.00	0.00%	14	124	1.56%
MCD250103P00290000	12/10/2024 8:22 PM	2025-01-03	2.00	0.00	0.00	0.00	0.00%	17	40	1.56%
MCD250110P00290000	12/10/2024 8:38 PM	2025-01-10	2.72	0.00	0.00	0.00	0.00%	6	12	1.56%
MCD250117P00290000	12/10/2024 8:53 PM	2025-01-17	3.60	0.00	0.00	0.00	0.00%	53	2,458	1.56%
MCD250221P00290000	12/10/2024 8:38 PM	2025-02-21	6.25	0.00	0.00	0.00	0.00%	7	134	0.78%
MCD250321P00290000	12/10/2024 8:56 PM	2025-03-21	8.11	0.00	0.00	0.00	0.00%	7	553	0.78%
MCD250620P00290000	12/10/2024 8:01 PM	2025-06-20	11.34	0.00	0.00	0.00	0.00%	9	304	0.78%
MCD250918P00290000	12/10/2024 8:43 PM	2025-09-18	14.61	0.00	0.00	0.00	0.00%	15	91	0.39%
MCD260116P00290000	12/6/2024 4:22 PM	2026-01-16	17.80	0.00	0.00	0.00	0.00%	34	157	0.39%
MCD261218P00290000	12/5/2024 5:45 PM	2026-12-18	26.00	0.00	0.00	0.00	0.00%	1	39	0.39%

301.42 +0.71 (+0.24%) 302.42 +1.71 (+0.57%)
At close: 9:31:06 AM EST Pre-Market: 9:29:55 AM EST €

Dec 13, 2024 300.00 List All Options

Calls

In The Money

Contract Name	Last Trade Date (EDT)	Expire Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213C003000000	12/10/2024 8:58 PM	2024-12-13	2.83	0.00	0.00	0.00	0.00%	2,399	1,931	0.00%
MCD241220C003000000	12/10/2024 8:55 PM	2024-12-20	5.05	0.00	0.00	0.00	0.00%	510	3,319	0.00%
MCD241227C003000000	12/10/2024 8:53 PM	2024-12-27	5.80	0.00	0.00	0.00	0.00%	194	403	0.00%
MCD250103C003000000	12/10/2024 8:39 PM	2025-01-03	6.80	0.00	0.00	0.00	0.00%	86	143	0.00%
MCD250117C003000000	12/10/2024 8:36 PM	2025-01-17	7.85	0.00	0.00	0.00	0.00%	14	43	0.00%
MCD250117C003000000	12/10/2024 8:58 PM	2025-01-17	7.82	0.00	0.00	0.00	0.00%	390	4,036	0.00%
MCD250124C003000000	12/10/2024 8:17 PM	2025-01-24	9.75	0.00	0.00	0.00	0.00%	22	23	0.00%
MCD250221C003000000	12/10/2024 8:46 PM	2025-02-21	12.86	0.00	0.00	0.00	0.00%	62	323	0.00%
MCD250321C003000000	12/10/2024 7:51 PM	2025-03-21	14.48	0.00	0.00	0.00	0.00%	139	1,571	0.00%
MCD250620C003000000	12/10/2024 6:53 PM	2025-06-20	19.27	0.00	0.00	0.00	0.00%	10	2,233	0.00%
MCD250919C003000000	12/10/2024 8:43 PM	2025-09-19	24.46	0.00	0.00	0.00	0.00%	3	217	0.00%
MCD260116C003000000	12/10/2024 7:32 PM	2026-01-16	29.08	0.00	0.00	0.00	0.00%	23	903	0.00%
MCD260116C003000000	12/6/2024 2:58 PM	2026-01-16	41.29	0.00	0.00	0.00	0.00%	2	155	0.00%
MCD270115C003000000	12/10/2024 3:05 PM	2027-01-15	38.40	0.00	0.00	0.00	0.00%	1	63	0.00%

301.42 +0.71 (+0.24%) 302.42 +1.71 (+0.57%)
At close: 9:31:06 AM EST Pre-Market: 9:29:55 AM EST €

Dec 13, 2024 300.00 List All Options

Puts

In The Money

Contract Name	Last Trade Date (EDT)	Expire Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213P003000000	12/10/2024 8:58 PM	2024-12-13	1.54	0.00	0.00	0.00	0.00%	144	245	0.78%
MCD241220P003000000	12/10/2024 8:57 PM	2024-12-20	3.10	0.00	0.00	0.00	0.00%	89	549	0.39%
MCD241227P003000000	12/10/2024 8:59 PM	2024-12-27	3.50	0.00	0.00	0.00	0.00%	94	80	0.39%
MCD250103P003000000	12/10/2024 8:32 PM	2025-01-03	3.60	0.00	0.00	0.00	0.00%	31	15	0.39%
MCD250117P003000000	12/9/2024 7:34 PM	2025-01-17	4.53	0.00	0.00	0.00	0.00%	1	7	0.30%
MCD250117P003000000	12/10/2024 8:58 PM	2025-01-17	5.47	0.00	0.00	0.00	0.00%	111	2,889	0.30%
MCD250221P003000000	12/10/2024 7:27 PM	2025-02-21	8.16	0.00	0.00	0.00	0.00%	21	110	0.30%
MCD250321P003000000	12/10/2024 4:30 PM	2025-03-21	11.10	0.00	0.00	0.00	0.00%	25	1,037	0.10%
MCD250620P003000000	12/9/2024 3:14 PM	2025-06-20	14.50	0.00	0.00	0.00	0.00%	42	320	0.10%
MCD250919P003000000	12/6/2024 7:43 PM	2025-09-19	17.10	0.00	0.00	0.00	0.00%	8	211	0.10%
MCD260116P003000000	12/10/2024 7:32 PM	2026-01-16	19.82	0.00	0.00	0.00	0.00%	27	345	0.10%
MCD260116P003000000	11/27/2024 3:05 PM	2026-01-16	28.50	0.00	0.00	0.00	0.00%	1	288	0.08%

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Calls

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Contract Name	Last Trade Date (EDT)	Expire Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213C003100000	12/10/2024 8:59 PM	2024-12-13	0.18	0.00	0.00	0.00	0.00%	475	917	6.25%
MCD241220C003100000	12/10/2024 8:57 PM	2024-12-20	0.87	0.00	0.00	0.00	0.00%	242	2,128	3.13%
MCD241227C003100000	12/10/2024 8:54 PM	2024-12-27	1.32	0.00	0.00	0.00	0.00%	69	396	3.13%
MCD250103C003100000	12/10/2024 8:57 PM	2025-01-03	1.90	0.00	0.00	0.00	0.00%	38	258	3.13%
MCD250110C003100000	12/10/2024 8:59 PM	2025-01-10	2.81	0.00	0.00	0.00	0.00%	19	68	3.13%
MCD250117C003100000	12/10/2024 8:57 PM	2025-01-17	3.34	0.00	0.00	0.00	0.00%	470	3,281	1.56%
MCD250124C003100000	12/10/2024 8:01 PM	2025-01-24	4.75	0.00	0.00	0.00	0.00%	2	10	1.56%
MCD250221C003100000	12/10/2024 8:57 PM	2025-02-21	7.25	0.00	0.00	0.00	0.00%	11	222	1.56%
MCD250321C003100000	12/10/2024 8:59 PM	2025-03-21	8.70	0.00	0.00	0.00	0.00%	80	830	1.56%
MCD250620C003100000	12/10/2024 8:21 PM	2025-06-20	14.77	0.00	0.00	0.00	0.00%	5	651	0.78%
MCD250919C003100000	12/10/2024 5:59 PM	2025-09-19	18.47	0.00	0.00	0.00	0.00%	1	97	0.78%
MCD260116C003100000	12/10/2024 5:33 PM	2026-01-16	23.40	0.00	0.00	0.00	0.00%	4	149	0.78%

301.42 +0.71 (+0.24%) 302.42 +1.71 (+0.57%)
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MCD241213C003050000	12/10/2024 8:58 PM	2024-12-13	0.72	0.00	0.00	0.00	0.00%	1,173	867	3.13%
MCD241220C003050000	12/10/2024 8:57 PM	2024-12-20	2.00	0.00	0.00	0.00	0.00%	1,330	2,257	1.56%
MCD241227C003050000	12/10/2024 8:39 PM	2024-12-27	3.10	0.00	0.00	0.00	0.00%	134	279	1.56%
MCD250103C003050000	12/10/2024 8:59 PM	2025-01-03	3.55	0.00	0.00	0.00	0.00%	188	164	1.56%
MCD250110C003050000	12/10/2024 8:36 PM	2025-01-10	4.96	0.00	0.00	0.00	0.00%	26	54	1.56%
MCD250117C003050000	12/10/2024 8:59 PM	2025-01-17	5.37	0.00	0.00	0.00	0.00%	528	2,976	0.78%
MCD250124C003050000	12/10/2024 8:44 PM	2025-01-24	6.63	0.00	0.00	0.00	0.00%	6	10	0.78%
MCD250221C003050000	12/10/2024 8:48 PM	2025-02-21	10.00	0.00	0.00	0.00	0.00%	25	146	0.78%
MCD250321C003050000	12/10/2024 8:07 PM	2025-03-21	11.85	0.00	0.00	0.00	0.00%	17	730	0.78%
MCD250620C003050000	12/10/2024 8:14 PM	2025-06-20	17.25	0.00	0.00	0.00	0.00%	5	405	0.39%
MCD250919C003050000	12/10/2024 8:14 PM	2025-09-19	21.90	0.00	0.00	0.00	0.00%	1	69	0.39%
MCD260116C003050000	12/9/2024 8:34 PM	2026-01-16	26.10	0.00	0.00	0.00	0.00%	4	570	0.39%
MCD261218C003050000	11/1/2024 3:33 PM	2026-12-18	35.75	36.30	38.50	0.00	0.00%	2	30	23.73%

301.42 +0.71 (+0.24%) 302.42 +1.71 (+0.57%)
At close: 9:31:06 AM EST Pre-Market: 9:29:55 AM EST €

Dec 13, 2024 315.00 List All Options

Calls

In The Money

Contract Name	Last Trade Date (EDT)	Expire Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
MCD241213C003150000	12/10/2024 8:37 PM	2024-12-13	0.07	0.00	0.00	0.00	0.00%	50	423	12.50%
MCD241220C003150000	12/10/2024 8:49 PM	2024-12-20	0.31	0.00	0.00	0.00	0.00%	165	1,506	6.25%
MCD241227C003150000	12/10/2024 8:32 PM	2024-12-27	0.73	0.00	0.00	0.00	0.00%	14	386	6.25%
MCD250103C003150000	12/10/2024 8:35 PM	2025-01-03	1.17	0.00	0.00	0.00	0.00%	30	235	3.13%
MCD250110C003150000	12/10/2024 8:52 PM	2025-01-10	1.50	0.00	0.00	0.00	0.00%	16	29	3.13%
MCD250117C003150000	12/10/2024 8:59 PM	2025-01-17	2.08	0.00	0.00	0.00	0.00%	223	2,100	3.13%
MCD250221C003150000	12/10/2024 7:45 PM	2025-02-21	6.05	0.00	0.00	0.00	0.00%	12	7	3.13%
MCD250321C003150000	12/10/2024 8:30 PM	2025-03-21	7.50	0.00	0.00	0.00	0.00%	20	1,221	1.56%
MCD250620C003150000	12/9/2024 6:02 PM	2025-06-20	12.20	0.00	0.00	0.00	0.00%	6	268	1.56%
MCD250919C003150000	12/9/2024 7:51 PM	2025-09-19	16.50	0.00	0.00	0.00	0.00%	1	37	1.56%
MCD260116C003150000	12/9/2024 8:46 PM	2026-01-16	21.10	0.00	0.00	0.00	0.00%	4	1,010	0.78%

Product 1: Partially Principal Protected Note

Below shows the parameters we need from real market:

- Current stock price: $S_0 = 301,76$
- Time to maturity: $T = 1,09$ (from 11-12-24 to 16-01-26)
- Annual risk-free interest rate: $r = 4,04\%$
- Annual volatility: $\sigma = 11,15\%$
- $N = 100$

where we assume the risk-free interest rate (R_f) is the rate of the 52- week US Treasury Bill on bank discount, currently standing at 4,04%, and we obtain volatility and other market information from Yahoo Finance.

Also, we predict the S_T , which is the stock price on January 16, 2026, using the Binomial Tree. To use Binomial Tree, we assume that:

- The underlying asset's price can either increase or decrease by a factor of u or d , respectively, over a given period.
- The probability of the underlying asset increasing or decreasing by a factor of u or d is calculated using the risk-neutral probability.

We divide the total time T into 4 equal steps, the details codes are run in R. The results are:

S_T	Probability
377,70	0,11
336,04	0,33

298,98	0,36
266,00	0,17
236,66	0,03

Construct the structured product:

From the bank's perspective, the primary costs involve the initial outlay required to construct the structured product. This includes purchasing a zero-coupon bond or other fixed-income instruments to ensure the promised principal protection at maturity, as well as acquiring the necessary options to replicate the product's payoff structure. As a result, if the formula below is larger than the 0, it means that the banks can earn from it.

$$N - \frac{0.9N}{(1 + R_f)^T} - \left(\frac{N}{S_0} \times \text{put price} \right) - \left(\frac{N}{S_0} \times \text{call price} \right) > 0$$

For the investors, the benefit is receiving at least 90% of the initial principal and possibly more if the underlying price moves favorably. The investor's expected outcome includes a baseline of significant principal protection plus a chance for enhanced returns, offering a balanced trade-off between safety and potential gain. **More calculations and explications are in Excel.**

The following table exhibits the corresponding payoff for investors and banks under different circumstances.

ST	377,70	336,04	298,98	266,00	236,66		
	Payoff of Investors	Payoff of Investors	Payoff of Investors	Payoff of Investors	Payoff of Investors	Expected Pyaoff	Profit Margin of banks
K1=280 K2=305	122,38	108,57	90,00	94,64	104,36	101,02	0,53
K1=290 K2=310	119,06	105,26	90,00	97,95	107,67	100,23	0,86
K1=295 K2=315	117,41	103,60	90,00	99,61	109,33	99,83	0,94

Descriptive part on Product 1: Partially Principal Protected Note

Key Features:

1. **Principal Protection:** Investors are guaranteed to receive at least 90% of their initial investment at maturity, ensuring minimal risk to their capital.
2. **Upside Potential:** Additional returns are tied to the performance of McDonald's Corporation (MCD) stock. Depending on the stock's maturity price (ST), investors may receive substantial premiums.
3. **Tailored Premiums:** The structured payoff ensures fair compensation in both favorable and moderate market scenarios. Payoffs are structured to provide returns even when the market doesn't perform optimally.

Why Invest:

1. **Safety with Growth:** Combines the security of principal protection with the potential for attractive returns.
2. **Optimized Payoffs:** Premium calculations are derived using real-time data and the binomial tree model, providing precise projections.

Illustrative Payoff Table:

1. The payoff structure is detailed in a table, showing different scenarios of McDonald's stock price (ST) at maturity. Expected payoffs and profit margins are calculated to showcase balanced returns for both investors and the bank.

Product 2: ATM digital

In this section, the additional premium $m \cdot N$ essentially represents the payoff of a digital call option struck at S_0 , paying out $m \cdot N$ if $S_T \geq S_0$. To fairly price this product, we need to consider the cost of ensuring full return of N at maturity (which can be done by investing part of N in a zero-coupon bond to guarantee principal repayment).

The bank's goal is to set m such that: The initial proceeds (N) are enough to cover the cost of the zero-coupon bond ensuring principal protection and the cost of purchasing the digital call option that replicates $m \cdot N$ payout if $S_T \geq S_0$. After covering these costs, a margin or profit for the bank remains.

Determine for m :

The assumptions are the same as in Product 1. Also, let's denote the fair value of the digital call that pays 1 unit as D . If we need the digital call to pay m (a multiplier on the notional N), the cost would be $m \cdot N \cdot D$ (since D is the cost for a payout of 1 unit, and we need $m \cdot N$ units). Hence:

$$N = PV_{bond} + (m \cdot N \cdot D)$$

solve for m :

$$m = (N - PV_{bond}) / (D \cdot N)$$

Determine for D :

Next step is to determine N . Normally, we have three ways to calculate it.

1. Using the Black-Scholes to determine the digital call option price.
2. Approximating the Digital Call Using a Tight Call Spread

First, we use the **Black-Scholes to determine the digital call option price**. Thus, we assume that:

1. The price of the underlying asset follows a geometric Brownian motion under the risk-neutral measure, with constant drift equal to the risk-free rate and constant volatility.
2. The market is frictionless, and investors can trade the assets and hedge continuously without any liquidity issues.
3. No Arbitrage and Unlimited Borrowing/Lending at the Risk-Free Rate:
4. There are no arbitrage opportunities, and investors can borrow and lend money at the constant risk-free interest rate.

We know that for European Call Option Pricing Formula (No Dividends), the price can be determined as:

$$EC(K, T) = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

where:

$$d_1 = (\ln(S_0/K) + (r + \frac{\sigma^2}{2})T) / \sigma\sqrt{T}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Here, $N(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution.

From the risk-neutral pricing theory, $N(d_2)$ represents the probability under the risk-neutral measure that the underlying asset price at maturity, $S_T \geq K$, exceeds the strike price K , i.e.,

$$N(d_2) = Q(S_T \geq K).$$

A digital call option (cash-or-nothing call) pays \$1 if $S_T \geq K$, and \$0 otherwise. Thus, its present value is the probability of this event multiplied by the discount factor.

Thus, the price of a digital call option can be expressed as:

$$\text{Digital Call Price} = e^{-rT} \cdot N(d_2)$$

By applying the same assumption as above such that:

1. Current stock price: $S_0 = 301,76$
2. Time to maturity: $T = 1.09$ (from 11-12-24 to 16-01-26)
3. Annual risk-free interest rate: $r = 4.04\%$
4. Annual volatility: $\sigma = 11,15\%$
5. $N = 100$

we get the *Digital Call* = 0,6

Second, we use digital call using a tight call spread to determine the digital call option price. I have market prices for European vanilla calls at similar strikes, so I can approximate a digital call by constructing a very narrow call spread around K .

$$\text{Digital Call} \approx \frac{C(K) - C(K + \Delta K)}{\Delta K}$$

We obtain what we need from the real market, showing that $C(300) = 29,08$ (the call price at strike $K = 300$) and $C(305) = 26.10$ (the call price at strike $K = 305$), and the strike difference $\Delta K = 305 - 300 = 5$. Using the tight call spread approximation, we get that *Digital Call* = 0,6, same number as we get using BS models. Then we can calculate the **m** , which equals to 0,08, where banks earn zero profit.

Adjusting m Downward to Increase Profit:

If bank wants margin (profit), we should choose m slightly lower than the break-even computed above. The break-even m assumes zero profit. By setting m slightly lower, we reduce the digital payoff cost we must fund. This way, after covering bond and option costs, some part of the initial N remains as the bank's profit.

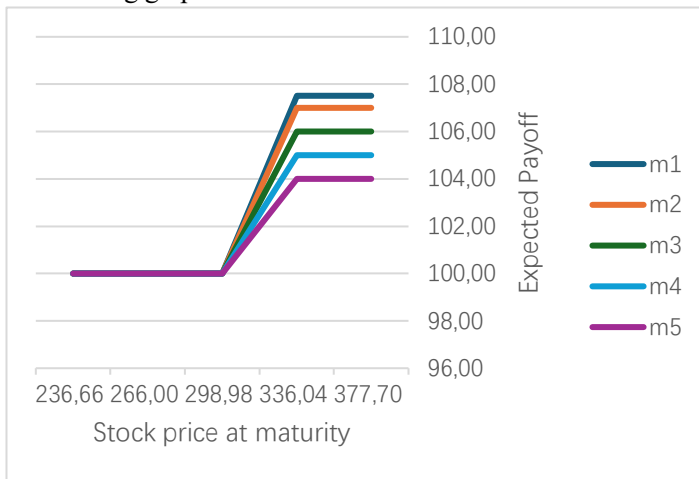
For example, here we decrease the m 4 times, from 0,08 to 0,04. **The changed payoff and expected payoff are shown in Excel.**

$m_2 = 0,07$: A slight reduction that may provide a small profit while still offering investors a potentially appealing upside.

$m_3 = 0,06$: A further reduction that increases the bank's margin but reduces investor's upside somewhat.

$m_4 = 0,05$ and below : A more conservative m , securing more profit for the bank but possibly making the product less attractive compared to alternatives in the market.

The following graph shows the results under different circumstances:



Results	Expected Payoff	Profit Margin of Bank
m1	103,32	0,00
m2	103,10	0,30
m3	102,65	0,89
m4	102,21	1,48
m5	101,77	2,07

Descriptive part on Product 2: ATM digital

Key Features:

- Simple and Transparent:** Investors earn a fixed multiplier on their initial investment if the stock price exceeds its current level at maturity.
- Full Principal Protection:** In less favorable scenarios, investors retain their entire initial investment.
- Competitive Multiplier:** The multiplier (m) is optimized using the Black-Scholes model, ensuring a fair and appealing return.

Why Invest:

- Low-Risk, High Certainty:** Investors gain upside exposure while safeguarding their principal.
- Efficient Pricing:** The product's design minimizes costs for investors and ensures a modest margin for the bank, maintaining overall attractiveness.
- Market Insights:** Based on robust calculations and market data, this product leverages predictive analytics for pricing.

Illustrative Payoff Table:

- Provides clear examples of expected payoffs under different market scenarios, ensuring investors understand potential returns.

Investor Benefits

- Tailored Risk Management:** Both products are designed with different risk appetites in mind, ensuring suitability for a range of investors.
- Transparency and Reliability:** Real-time market data and proven financial models underline the accuracy and fairness of pricing.

Sources:

- Risk-free rate:

https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_bill_rates&field_tdr_date_value=2024

- Market price data (Option prices, volatility, current stock price): Yahoo Finance